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Évaluation du niveau osseux résiduel autour des implants explantés. Facteurs locaux et systémiques

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Division de médecine dentaire régénérative
et de parodontologie

Thèse préparée sous la direction de la Professeure Catherine Giannopoulou

" Évaluation du niveau osseux résiduel autour des implants explantés. Facteurs locaux et systémiques "

Thèse
présentée à la Faculté de Médecine
de l'Université de Genève
pour obtenir le grade de Docteur en Médecine Dentaire
par

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De
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Évaluation du niveau osseux résiduel autour des implants explantés. Facteurs locaux et systémiques

La Faculté de médecine, sur le préavis du Comité directeur des thèses, autorise l'impression de la présente thèse, sans prétendre par-là émettre d'opinion sur les propositions qui y sont énoncées.

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Doyen



N.B. - La thèse doit porter la déclaration précédente et remplir les conditions énumérées dans les "Informations relatives à la présentation des thèses de doctorat à l'Université de Genève".

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RÉSUMÉ

OBJECTIFS :

Cette étude visait à évaluer le niveau osseux résiduel des implants explantés et à rapporter les caractéristiques locales et systémiques observées chez les patients ayant subi une explantation implantaire.

MATERIEL ET METHODES :

Un total de 494 dossiers administratifs a été initialement extrait de la base de données. Après élimination des doublons, 464 dossiers médicaux de patients ayant subi une explantation implantaire entre 2005 et 2021 dans treize cliniques privées en Suisse ont été examinés. Les informations concernant l'historique implantaire, les caractéristiques locales (santé bucco-dentaire, statut parodontal et prothétique) et systémiques (maladies générales, médication et tabagisme) ont été recueillies à partir des dossiers médicaux et des données radiographiques. Registre ISRCTN (ISRCTN10631004).

RESULTATS :

Au total, 399 patients et 521 implants ont été inclus dans l'analyse. La péri-implantite a été la cause la plus fréquente d'explantation (62 %). L'explantation a été principalement réalisée dans des cas de péri-implantite avancée, avec une perte osseuse moyenne de 62,92 % de la longueur implantaire. La majorité des patients présentaient une perte osseuse radiographique autour de la dentition restante avec un ratio perte osseuse/âge de 0,74, une santé bucco-dentaire compromise, et 47,80 % d'entre eux étaient fumeurs. De plus, seulement 56,12 % des patients ont déclaré être en bonne santé générale.

CONCLUSION

Les explantations implantaires étaient généralement réalisées lorsque la perte osseuse atteignait environ 60 % de la longueur de l'implant. Elles étaient le plus souvent observées dans les cas avancés de péri-implantite et chez les patients présentant un profil à haut risque de dégradation parodontale, tel qu'un antécédent de maladie parodontale ou une forte consommation de tabac.

INTRODUCTION

Les implants dentaires permettent le remplacement des dents manquantes depuis plus de 50 ans, avec une taux de réussite de 95% après 10 ans de suivi (Buser et al., 2017). On estime qu'environ 12 millions d'implants sont posés chaque année dans le monde (Albrektsson et al., 2014), la prévalence de patients avec implants dentaires étant supérieur à 5% en 2016 (Elani et al., 2018). Malgré ces résultats encourageants, les implants ne sont pas exempts de complications. La surface de l'implant n'étant pas exfoliable, le biofilm bactérien peut s'accumuler, entraînant une inflammation des tissus péri-implantaires (Pontoriero et al., 1994). Les maladies péri-implantaires se classent selon l'atteinte limitée à la muqueuse sans perte de l'os (**mucosite**), ou étendue avec une perte progressive de l'os péri-implantaire (**péri-implantite**) (Berglundh et al., 2018). Ces maladies ont une haute prévalence, la mucosite affectant environ 30% des implants et la péri-implantite environ 10% (Lee et al., 2017).

Lors de la pose d'un implant dentaire, la colonisation bactérienne se produit dans les 30 premiers minutes après la pose (Furst et al., 2007), et le biofilm se développe de manière similaire à celui autour des dents, avec la formation d'une flore complexe après une semaine (Quirynen et al., 2005). En l'absence de brossage, cette colonisation bactérienne entraîne une inflammation des tissus péri-implantaires (Meyer et al., 2017; Salvi et al., 2012) (**Figure 1**).

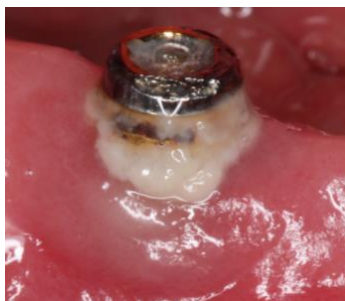


Figure 1. Inflammation de la muqueuse péri-implantaire en présence de plaque.

La relation cause-effet entre la plaque bactérienne et la mucosite a été montrée dans une étude menée chez 20 patients présentant une situation parodontale assainie et stable

pendant 6 mois après la pose implantaire. Ces patients ont interrompu le brossage pendant trois semaines pour permettre l'accumulation de la plaque. Ceci s'est traduit par une augmentation dans l'indice de saignement tant au niveau parodontal qu'au niveau péri-implantaire, montrant ainsi une relation cause-effet entre l'accumulation de la plaque et le développement de la mucosite (Pontoriero et al., 1994) (**Figure 2**).

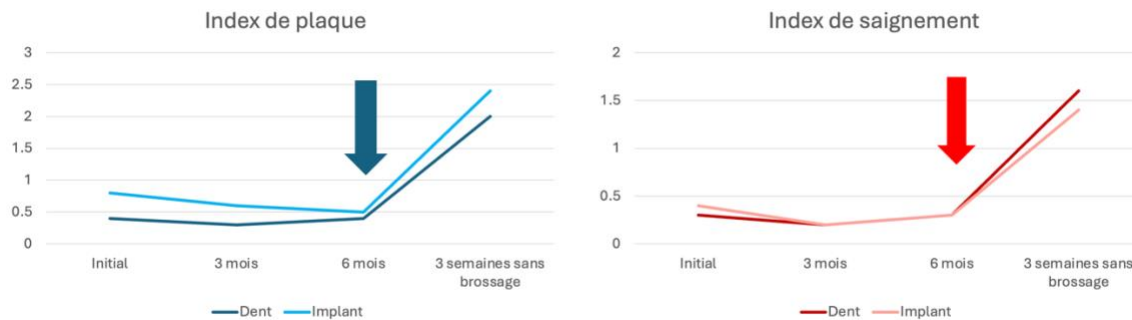


Figure 2. Représentation graphique des résultats de l'étude de Pontoriero, 1994. Les flèches indiquent le moment de l'interruption du brossage.

Peu d'études ont pu mettre en évidence une conversion de la mucosite en péri-implantite en raison de considérations éthiques. Toutefois, une étude a procédé à une réévaluation de patients qui avaient été diagnostiqués avec une mucosite cinq ans auparavant (Costa et al., 2012). Ces patients ont été recontactés et repartis en deux groupes : un groupe avec suivi (au moins un rendez-vous de suivi par an) et un groupe sans suivi. Les résultats ont montré des proportions inégales de péri-implantite entre les deux groupes (suivi : 18 % ; sans suivi : 44 %), mettant en évidence qu'en absence de suivi, le risque de conversion des mucosites en péri-implantites augmente. Les déterminants de cette conversion ne sont pas clarifiés, mais l'état parodontal (saignement au sondage, profondeur de sondage et présence de parodontite) semble être associé au développement de la péri-implantite.

La péri-implantite est une maladie d'origine bactérienne affectant les tissus environnants des implants dentaires, caractérisée par une inflammation de la muqueuse péri-implantaire et une perte osseuse progressive (Berglundh et al., 2018).

En clinique, le diagnostic se fait par une combinaison de plusieurs paramètres (Berglundh et al., 2018; Renvert et al., 2018) (**Figure 3**) :

- Inflammation muqueuse avec saignement au sondage et/ou suppuration
- Augmentation de la profondeur de sondage par rapport à celle obtenue après la mise en place de la suprastructure.
- Perte osseuse progressive par rapport au niveau osseux après 1 an de la mise en bouche de la reconstruction prothétique.
- Dans le cas où le sondage ou les radiographies initiales ne sont pas présentes, une perte osseuse péri-implantaire ≥ 3 mm et un sondage péri-implantaire ≥ 6 mm avec saignement.

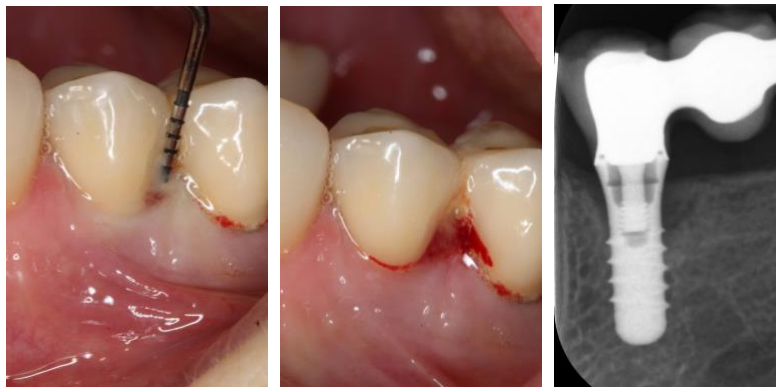


Figure 3. Exemple de péri-implantite avec sondage ≥ 6 mm, saignement, suppuration, et une perte osseuse radiographique ≥ 3 mm.

La péri-implantite présente une incidence dès les premières années de fonction de l'implant, avec une progression rapide et accélérée (Derks et al., 2016). Ainsi la perte osseuse progressive peut aboutir à la perte de l'implant, la péri-implantite étant la première cause d'explantation des implants dentaires, avec 64.5% des explantations réalisées par cette raison (Gargallo-Albiol et al., 2021; Monje & Nart, 2022). Ainsi, quand la perte osseuse produit une perte complète de l'ostéointégration, l'implant doit être explanté. Cependant, la décision d'explantation reste plus complexe dans les cas de péri-implantites sévères (Zimmer et al., 2025).

Quelques facteurs de risque ont été associés à un risque accru de péri-implantite, tels que l'antécédent de maladie parodontale, le manque de suivi et le mauvais contrôle de la plaque dentaire (Schwarz et al., 2018). Ces associations ont motivé les recommandations de la Fédération Européenne de Parodontologie pour la prévention des péri-implantites incluant l'évaluation du profil du risque du patient avec modification des facteurs et indicateurs de risque modifiables, le traitement préalable des gingivites et parodontites et l'établissement d'un programme de suivi et de maintenance avant la pose implantaire (Herrera et al., 2023). En effet, une étude a montré que seuls 9.2% des patients avaient reçu une information adéquate concernant les facteurs de risque liés aux implants (Insua et al., 2017).

Une fois établie, le traitement de la péri-implantite consiste en la décontamination de la surface implantaire. A l'instar du traitement des parodontites, la première phase commence par l'adaptation de l'hygiène bucco-dentaire du patient et de la prothèse, le sevrage tabagique, ainsi que la gestion des maladies concomitantes comme la parodontite et le diabète (Herrera et al., 2023). Ensuite le traitement se poursuit par un débridement péri-implantaire non chirurgical et une réévaluation entre 1.5 et 3 mois après le traitement. Lors de cette réévaluation, les poches supérieures à 5 mm avec saignement et/ou suppuration nécessitent une thérapeutique complémentaire, c'est-à-dire chirurgicale.

Dans la majorité des cas, notamment dans les cas avancés, le traitement non chirurgical n'est pas suffisant pour stabiliser la maladie, et le traitement chirurgical permet d'accéder à la surface de l'implant pour la décontamination grâce à un lambeau de pleine épaisseur (Renvert & Polyzois, 2018) (**Figure 4**).



Figure 4. Exemple de chirurgie péri-implantaire avec un lambeau d'accès. Situation initiale, décontamination implantaire et fermeture avec sutures. Réévaluation après 6 ans avec absence d'inflammation et stabilité des tissus.

Ces approches sont prévisibles et ont permis une réduction de l'inflammation, de la profondeur de sondage, ainsi qu'une stabilisation de la perte osseuse (Donos et al., 2023; Karlsson et al., 2023), avec un taux de survie d'environ 82 % des implants à 3 ans et 76 % à 5 ans (Roccuzzo et al., 2018). Il faut toutefois tenir compte d'un taux important de récurrences et de pertes implantaire pendant le suivi post-traitement, malgré la thérapie de soutien : entre 25% de cas (dont 11% avec explantation) (Heitz-Mayfield et al., 2018) et 44% (dont 21% avec explantations) (Carquac et al., 2020).

En effet, certaines situations cliniques ont montré des limites pour la stabilisation de la maladie péri-implantaire. Ainsi, la perte osseuse péri-implantaire initiale a été inversement corrélée au taux de réussite du traitement chirurgical et représente le critère pronostic principal (de Waal et al., 2016; Ravidà et al., 2022). D'autres facteurs comme le tabac (de Waal et al., 2016), l'état de surface implantaire (Carquac et al., 2020; Romandini et al., 2024), la profondeur de sondage initiale (Ichioka et al., 2023) le contrôle de plaque, l'observance du suivi, ou le diagnostic de parodontite sévère (Lagervall & Jansson, 2013) ont également montré une influence sur les résultats du traitement.

La perte osseuse initiale a été classifiée selon le type, le nombre de parois résiduelles et la sévérité (**Table 1**) (Monje et al., 2019).

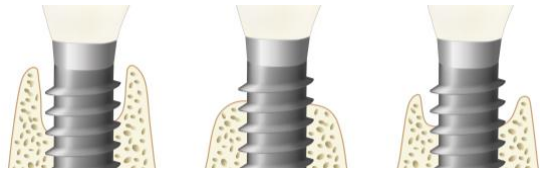
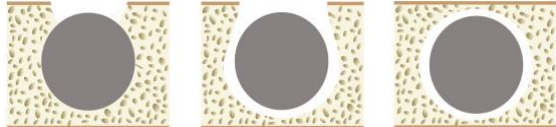
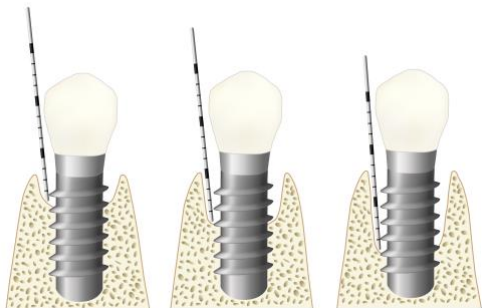
Niveau osseux résiduel	Classe I	Infra-osseuse	
	Classe II	Supra-osseuse	
	Classe III	Combinée	
Nombre de parois	Classe a	Déhiscence vestibulaire	
	Classe b	2 – 3 parois	
	Classe c	Circonférentiel - 4 parois	
Sévérité de la lésion	Grade S (Léger)	3 – 4 mm / < 25% implant	
	Grade M (Modéré)	4 – 5 mm / 25% - 50%	
	Grade A (Avancé)	> 6 mm / > 50%	

Table 1. Classification de la perte osseuse péri-implantaire selon Monje et al, 2019.

Ainsi les cas de parodontite sévère, avec une perte osseuse supérieure à 50% de la longueur de l'implant, ont montré une moins bonne réponse au traitement comparativement aux cas débutants (Ravida et al., 2022). En effet, une perte osseuse > 50% a été associée à une augmentation par 20 fois du risque de perte implantaire après le traitement, par rapport aux implants présentant une perte < 25% (Ravidà et al., 2020).

Ces résultats pourraient s'expliquer par le fait que les cas avancés présentent une surface implantaire exposée à la contamination bactérienne plus étendue, avec un risque accru de

bactéries résiduelles, notamment dans les surfaces rugueuses (Albouy et al., 2011). Dans les cas où le défaut est infra-osseux, la décontamination de la région implantaire proche de l'apex sera plus difficilement accessible (Steiger-Ronay et al., 2017) (**Figure 5**). De même, dans le cas où la perte osseuse est supra-osseuse, le repositionnement apical du lambeau pour la réduction des poches peut être limité par la profondeur du vestibule (Koldsland et al., 2018).

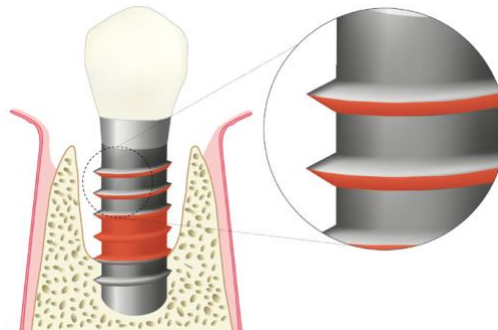


Figure 5. Régions de la surface de l'implant plus difficiles à décontaminer (en orange).

Actuellement, le seuil au-delà duquel l'explantation est indiquée provient davantage de l'opinion d'experts plus que de données scientifiques. Ainsi, une perte osseuse de 50% a été établie comme une limite au-delà de laquelle le pronostic devient moins prévisible (Zimmer et al., 2025). Une étude rétrospective sur implants explantés a montré qu'en clinique, les implants sont explantés en moyenne après une perte osseuse d'environ 66,2% de la longueur de l'implant (Wentorp et al., 2021). Toutefois, cette étude n'a pas pu analyser d'autres facteurs tels que les éléments prothétiques, ainsi que l'état dentaire et parodontal des patients (Wentorp et al., 2021).

Le but de cette recherche rétrospective est (1) évaluer le niveau de perte osseuse associé aux décisions d'explantation, et (2) caractériser les profils locaux et systémiques des patients ayant suivi ces explantations.

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Assessment of Residual Bone Level Around Explanted Implants and Description of the Systemic and Local Characteristics. A Retrospective Study

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Correspondence: Rodrigo Martin-Cabezas (rmartincabezas@gmail.com)**Received:** 6 March 2025 | **Revised:** 3 June 2025 | **Accepted:** 8 July 2025**Keywords:** explantation | marginal bone loss | peri-implantitis | treatment outcome

ABSTRACT

Objectives: The study aimed to evaluate the residual bone level of explanted implants and to report local and systemic characteristics observed in patients who underwent implant explantation.

Material and Methods: A total of 494 administrative records were initially retrieved from the database. After elimination of duplicates, 464 medical records of patients who underwent implant explantation between 2005 and 2021 in thirteen private clinics in Switzerland were screened. Information regarding implant history, local (oral health, periodontal and prosthetic status) and systemic (systemic diseases, medication, and smoking) characteristics was assessed through the medical records and the radiographic data. ISRCTN registry (ISRCTN10631004).

Results: In total, 399 patients and 521 implants were included in the analysis. Peri-implantitis was the most frequent reason for explantation (62%). Explantation was mainly performed in cases of advanced peri-implantitis with a mean bone loss of 62.92% of the implant length. Most patients had radiographic bone loss around the remaining dentition with a bone loss/age ratio of 0.74, presented compromised oral health, and 47.80% of them were smokers. Moreover, only 56.12% of patients reported being in good systemic health.

Conclusion: Implant explantations were typically performed once bone loss reached approximately 60% of the implant length. They were most often observed in advanced cases of peri-implantitis and in patients with high-risk profiles for periodontal breakdown, such as a history of periodontal disease or heavy smoking habits.

1 | Introduction

Peri-implantitis treatment has shown high survival rates of the affected implants (Roccuzzo et al. 2018); however, the outcomes of these interventions are less predictable than those for periodontal treatments, with recurrence rates reaching 45% after 5 years (Carcuac et al. 2020). In advanced cases, bone loss can lead to implant failure (Monje and Nart 2022). Consequently, one of the most significant prognostic factors in peri-implant treatment is the level of initial bone loss (de Waal et al. 2016). When bone loss exceeds 50% of the length of the implant,

explantation has been recommended (Monje and Nart 2022; Sinjab et al. 2018), as treatment beyond this threshold becomes 20 times less predictable compared to implants with bone loss <25% (Ravida et al. 2022).

Determining the residual bone level beyond which an implant cannot be preserved and explantation is recommended, remains challenging (Martin-Cabezas and Giannopoulou 2024). Recent studies suggest a threshold of two-thirds of the implant length (66.2%) as the point at which practitioners are more likely to proceed with extraction. However, limited scientific data exist on

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this topic, and larger cohorts are needed to analyze the influence of additional factors (Wentorp et al. 2021).

Some risk factors for peri-implantitis are well-established, such as history of periodontal disease, poor plaque control, or lack of follow-up, whereas others, such as diabetes and smoking, until recently remained inconclusive (Schwarz et al. 2018). The periodontal condition of adjacent teeth can influence the microbiological colonization of the implants (Eick et al. 2016), emphasizing the importance of controlling periodontal pockets in patients with implant-supported restorations. Additionally, periodontal bone loss relative to age is a critical factor for understanding the progression rate of periodontitis (Tonetti et al. 2018). This factor may also elucidate the history of periodontitis, which has been associated with peri-implantitis (Carra et al. 2022), particularly in severe cases (Roccuzzo et al. 2014) and in non-compliant patients (Roccuzzo et al. 2023). Moreover, prosthetic restorations can hinder access to plaque control (Serino and Strom 2009), and specific prosthetic factors, such as emergence angles, emergence profiles, or splinting, have been associated with increased peri-implantitis prevalence (Yi et al. 2020). Only recently, there is growing evidence of the influence of systemic diseases, such as hyperglycemia and smoking, on the prevalence of peri-implant diseases (Monje et al. 2023).

Given this context, the aim of this retrospective study is to analyze the residual bone level of implants at the time of explantation and to describe the reasons for this procedure. We further assessed the local (dental, periodontal, prosthetic and implant-related factors) and systemic (age, gender, systemic diseases, medications, and smoking) characteristics of patients who underwent implant explantation. By defining a patient profile for implant loss, this study aims to assist practitioners in strengthening monitoring and preventive treatment.

2 | Material and Methods

2.1 | Study Design and Setting

The data used in this retrospective study were extracted from patient records, identified from the billing database of a group of 13 private dental clinics in Switzerland between January 2005 and December 2021. The study was conducted in accordance with the Declaration of Helsinki on human studies, following approval from the Ethics Committee for Research of Geneva (Switzerland) under protocol reference number 2022-00151 for the use of health-related data without individual consent, in accordance with Swiss Law Art. 34 LRH, Art. 37-40 ORH. The study protocol was registered at the ISRCTN registry (ISRCTN10631004) and the results were reported according to the STROBE guidelines (Vandenbroucke et al. 2007).

2.2 | Participants

To be eligible for the study, the medical record had to include: (i) data related to adult patients with an explanted dental implant during the inclusion period; (ii) radiographic data at explantation; and (iii) the reason for explantation.

The exclusion criteria were: (i) unclear reason for explantation; (ii) incomplete radiographs (radiographs that do not allow measurement of the entire implant or are unusable due to exposure/distortion, etc.); (iii) implants placed for orthodontic reasons, needle-shaped implant, temporary implants; (iv) non-identifiable position of the explanted implant, or (v) document confirming patient's refusal.

2.3 | Collected Data

The medical records were screened, and the following data were collected when available:

- At the patient level: age at the time of explantation, gender, history of periodontal treatment, systemic conditions, medication use, and tobacco consumption.
- At the implant level: date of explantation, position of the implant, implant lifespan, type of implant, reason for explantation, initial periodontal diagnosis, periodontal probing depth at explantation, and frequency of follow-up prior to the explantation.

2.4 | Radiographic Assessment

Periapical, panoramic, and CBCT X-ray files were analyzed with the CS imaging software version 7.0.3 and 8 (Carestream Dental) by the same examiner (R.M.C.) who assessed the following:

- At the patient level: number of missing teeth, number of implants, the decayed, missing, and filled teeth (DMFT) index, bone loss at teeth, and ratio bone loss/age.
- At the implant level: marginal bone loss at explantation (mm) from the implant shoulder to the most apical part of the bone defect, relative bone loss at explantation (%) assessed as percentage of the implant length with peri-implant bone loss (Figure 1), type of the peri-implant bone defect as follows: class I (intraosseous), class II (supracrestal/horizontal) or class III (combined) and severity of the defect as follows: Grade S (Slight: 3–4 mm/< 25% of the implant length), Grade M (Moderate: 4–5 mm/≥ 25%–50%) or Grade A (Advanced: > 6 mm/> 50%) (Monje et al. 2019). Additionally we recorded the type, extent and splinting of the prosthesis, number of abutments, position of the implant on the prosthetic restoration, presence and position of a cantilever, presence of micro-gap between the implant and the prosthesis, emergence angle and profile at mesial and distal as defined by the Glossary of Prosthodontic Terms, 9th edition ("The Glossary of Prosthodontic Terms: Ninth Edition," 2017), crown-implant ratio and type of antagonist and proximity to neighboring anatomical structures (tooth/implant, sinus, nasal cavity, inferior alveolar nerve).

Intra-rater reliability was not formally assessed in this study. However, all radiographic evaluations were performed by a single experienced examiner following a standardized protocol to minimize variability.

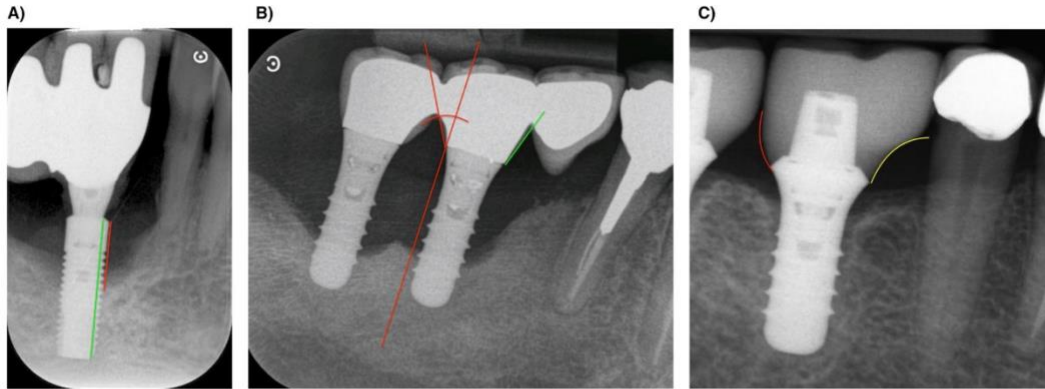


FIGURE 1 | Assessment of radiographic data. (A) The bone loss was assessed in millimeters from the shoulder of the implant to the apical part of the bone defect (red line). The relative bone loss was reported as a percentage of the total implant length (green line) affected by the defect. (B) Emergence angle of the prosthetic reconstruction measured as the angle between the tangent of the transitional contour relative to the long axis of the implant (red lines). (B and C) Emergence profile of the prosthetic restoration was classified as droit (green line), convex (red line), or concave (yellow line).

2.5 | Variables and Outcomes

To achieve the study's objectives, we focused on several clinical and radiographic variables. The primary objectives were to evaluate: (i) the amount of residual bone level around implants at the time of explantation (Outcome 1) and the prevalence of peri-implantitis as the main reason for explantation (Outcome 2).

Furthermore, we describe several local characteristics (implant-related factors, periodontal and dental factors, prosthetic reconstruction factors) and systemic characteristics related to the patient (systemic conditions, medications, tobacco consumption) (Outcome 3 and 4, respectively).

2.6 | Statistical Analysis

The entire cohort was analyzed for explantation reasons, and further sub-analysis was performed only on the patients who underwent an explantation due to peri-implantitis. Quantitative variables were summarized using descriptive statistics, including means, standard deviations (SD), averages, and ranges, depending on the variable distribution. Categorical data were expressed as frequencies and percentages. When some data was missing, the patient or the implant concerned was not included in that specific analysis. The sample size for each analysis is, in consequence, reported on each table.

3 | Results

3.1 | Participants

A total of 494 patient files were identified as having undergone implant explantations through the billing database. After resolving duplicates ($n=30$), which included cases where the same patient was treated in multiple clinics, a total of 464 medical files corresponding to 609 implants were screened. Following exclusions,

399 patients and 521 implants were included in the analysis. Figure 2 presents the study inclusion flowchart, and Table 1 details the reasons for exclusion.

3.2 | Descriptive Data

Demographic characteristics of the included population (all reasons of explantation included) and the subset with peri-implantitis-related explantations are presented in Table 2. The distribution of explantations was equal between both sexes. The age at the time of explantation ranged from 17 to 88 years, with a mean age of 60.40 years. Only one patient was younger than 20 years old, and in this case, the implant was extracted due to primary failure. Some implants were removed within the first year after placement, whereas others remained functional for more than 35 years, with a mean implant lifespan of 5.14 years. As shown in Figure 3 the survival rate dropped sharply within the first 6 years, followed by a more gradual decrease. Explanted implants were more frequently located in the maxilla than in the mandible (58.51% and 41.49%, respectively) and posterior sextants were more commonly affected than anterior sites.

When the number of explantations per patient was analyzed, the mean value was 1.31. Although most patients (80.95%) underwent only one explantation, some had multiple implants removed: two explantations (13.03%); 3 (3.01%); 4 (2%) or more (1%). A few individuals experienced a cluster of explantations, with more than five implants removed.

3.3 | Main Results

3.3.1 | Outcome 1. Bone Level at Explantation Time

The residual bone level at the time of explantation was analyzed only for the peri-implantitis patients, and after exclusion of implants presenting 100% bone loss, provided more

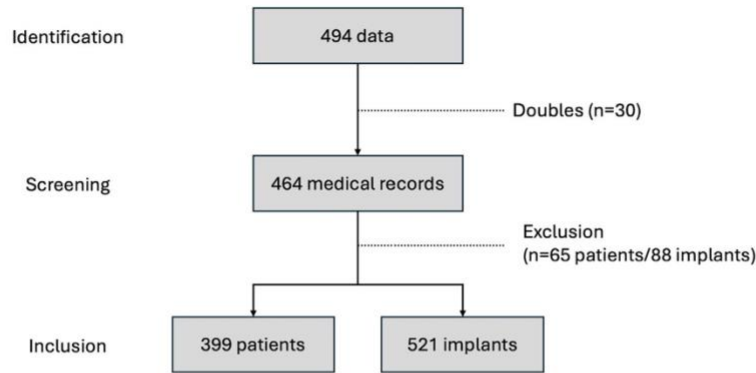


FIGURE 2 | Flow chart of study inclusion.

TABLE 1 | Reasons for exclusion.

	Patients	Implants
Orthodontic/needle-shaped implant or apex	31 (47.69%)	34 (36.56%)
Absence of X-rays	17 (26.15%)	24 (25.81%)
Reason for explantation no identifiable	5 (7.69%)	9 (9.68%)
X-rays no exploitable	2 (3.08%)	7 (7.53%)
Position of the implant no identifiable	4 (6.15%)	4 (4.30%)
Other reasons ^a	6 (9.23%)	10 (11.36%)

^aBilling code error ($n=2$), implant fell out on its own ($n=2$), explantation requested by the patient despite the absence of any clinical justification ($n=6$).

accurate data regarding the explantation decision. The type and severity of the defect were also classified according to Monje et al. (2019) (Monje et al. 2019). The relative bone loss ranged from 18.84% to 94.21% with a mean value of 62.92%, corresponding to 6.76 mm of bone loss. Most explantations (50.79%) involved combined supra- and infra-bony defects, and 80.42% of the peri-implantitis cases were classified with advanced severity (Table 3). The number of residual bone walls could not be assessed as only radiographs were available in the medical records.

Figure 4 shows three cases in which explantation was due to peri-implantitis (showing approximately 60% bone loss).

3.3.2 | Outcome 2. Prevalence of Peri-Implantitis Among the Reasons for Explantation

The explanations were classified based on their primary reason (Table 4). Peri-implantitis was the most reported reason, accounting for 62% of cases. This was followed by primary failure (16.51%), aseptic loosening (11.32%) and fractures (5.95%).

3.3.3 | Outcome 3. Local Characteristics: Implant, Prosthetic, and Dental/Periodontal Status

Different local factors were analyzed in the peri-implantitis group when data were available (Table 5). Implant-related factors revealed a mean diameter of 4.21 mm, with a low proportion of narrow implants (11.88%). Additionally, 30.21% of the implants had a length of less than 10 mm. Limited information was available regarding other factors, such as the manufacturer of the implant, implant surface, or surgical techniques, especially for patients referred externally. This limitation prevented a detailed analysis of these factors, as they might have been influenced by the local clinical practices.

At the time of explantation, some patients had lost all their natural teeth, whereas for others, the implant replaced only a single missing tooth (Table 6). The mean number of missing teeth in this population was 14.41, with an average of 4.74 implants per patient. The mean DMFT index was 22.75, and this value remained high when the 25 completely edentulous patients were excluded (mean DMFT index = 22.05). Radiographic analysis revealed that nearly every patient had clinical attachment loss, with bone loss averaging 45% of the root length. Furthermore, the bone loss/age ratio was 0.74, representing a moderate rate of periodontal progression (Tonetti et al. 2018). Despite this, only 18% of patients reported a history of periodontal treatment.

When initial periodontal charting was available ($N=52$), periodontitis was classified by stages (Papapanou et al. 2018): stage I (0%), stage II (2%), stage III (58%), and stage IV (40%), and by grades: grade A (0%), grade B (50%), and grade C (50%). However, the presence of deep pockets (>5 mm) at the time of explantation was limited to 17.33% of the patients.

Prosthetic restoration data highlighted a predominance of fixed prostheses (71.52%) compared to removable ones (18.89%). Natural teeth were the most common antagonists (48.30%), followed by implant-supported fixed prostheses (12.07%) and removable prostheses (10.22%) (Table 5). Analyzing the characteristics of fixed prostheses (Table 7), explantations were almost equally distributed between unitary (42.42%) and splinted restorations (57.58%). Poor adaptation between the prosthetic reconstruction and the

TABLE 2 | Demographic data of the included population.

	All explantations <i>n</i> (%/SD)	Explantations due to peri-implantitis <i>n</i> (%/SD)
Gender	<i>N</i> = 399 patients	<i>N</i> = 242 patients
Male	193 (48.37%)	112 (46.28%)
Female	206 (51.63%)	130 (53.72%)
Number of explantations per patient	<i>N</i> = 399 patients	<i>N</i> = 242 patients
Explantations per patient	1.31	1.33
Patients with 1 implant explanted	323 (80.95%)	191 (78.93%)
Patients with 2 implants explanted	52 (13.03%)	34 (14.05%)
Patients with 3 implants explanted	12 (3.01%)	8 (3.30%)
Patients with 4 implants explanted	8 (2.01%)	7 (2.89%)
Patients with 5 implants explanted	1 (0.25%)	0 (0%)
Patients with more than 5 implants explanted	3 (0.75%)	2 (0.83%)
Patient age at explantation (per implant) ^a	<i>N</i> = 521 implants	<i>N</i> = 323 implants
Years	60.40 (13.17)	60.87 (13.66)
Position of the implant	<i>N</i> = 521 implants	<i>N</i> = 323 implants
Maxillary	317 (60.84%)	189 (58.51%)
Mandibular	204 (39.15%)	134 (41.49%)
Sextant 1	105 (20.15%)	65 (20.12%)
Sextant 2	99 (19%)	58 (17.96%)
Sextant 3	113 (21.69%)	66 (20.43%)
Sextant 4	87 (16.7%)	58 (17.96%)
Sextant 5	41 (7.87%)	31 (9.60%)
Sextant 6	76 (14.59%)	45 (13.93%)
Implant lifespan	<i>N</i> = 337 implants	<i>N</i> = 173 implants
Years	5.14 (5.75)	7.15 (5.84)

^aFor patients with multiple explantations, age was recorded individually for each implant at the time of explantation.

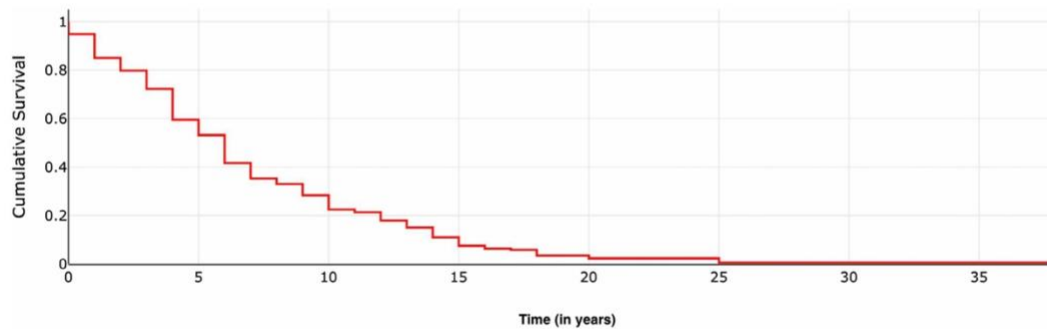


FIGURE 3 | Time to implant explantation: Survival analysis.

implant was observed in 24% of radiographs, and 12% of implants had a combined restoration supported by both teeth and implants. Although single-unit restorations were the most common (43%),

bridges with at least four crowns were the second most frequent (23.48%). Explantations were more prevalent in implants with distal cantilever or straight emergence profiles.

The characteristics of removable prostheses are detailed in Table 8. In two cases, the medical records did not provide sufficient information to determine whether the implant supported a complete or partial prosthesis. As a result, these two patients were excluded from the detailed analysis of prosthesis-related characteristics. There were 59 explanted implants with removable prostheses. Forty-nine were included in complete reconstructions and ten in partial prostheses. It was not possible to assess the number of abutments for four implants because the periapical X-rays and the medical file did not allow for all the abutments to be visualized. The analysis concerning the number of abutments was therefore performed on 55 implants. Most of the removed implants served as abutments for complete removable prostheses (83.05%), predominantly in the maxilla (71.43%). One-third of these implants were splinted. Regarding

the number of abutments, four was the most frequent configuration (34.55%). Importantly, no explantations occurred in mandibular sites with fewer than two implants, whereas 19.35% of implants were removed in maxillary positions with fewer than four implants. Lastly, a small proportion of implants (7%) served as abutments for combined restorations involving both teeth and implant-supported prostheses.

3.3.4 | Outcome 4. Systemic Characteristics: Systemic Conditions, Medication, and Tobacco Consumption

Systemic characteristics related to the patients' profile were analyzed in the peri-implantitis group and are described in Table 9. Regarding systemic health, 56.12% of patients reported being in good health. Certain conditions prevalent in more than 10% of the population included cardiovascular diseases, hypertension, and high cholesterol (14.77%, 22.78%, and 10.97%, respectively). Diabetes was present in 7.17% of patients.

Medication use corresponded to the reported health conditions. Antiplatelet agents, antihypertensive medications, and statins were used by more than 10% of patients, whereas antidiabetic medications were taken by 6.84% of patients.

TABLE 3 | Bone loss at the explantation time in the peri-implantitis group after exclusion of implants with bone loss = 100%.

	<i>n</i> (%)	SD	Max	Min
Bone loss at explantation (<i>N</i> = 192)				
Bone loss (mm)	6.76	2.09	12.2	1.9
Relative bone loss (%)	62.92	16.49	94.21	18.84
Type of defect (<i>N</i> = 192) ^a				
Class I	44 (23.28%)			
Class II	49 (25.93%)			
Class III	96 (50.79%)			
Severity of the defect (<i>N</i> = 192) ^a				
Slight	1 (0.53%)			
Moderate	36 (19.05%)			
Advanced	152 (80.42%)			

^aAs described by Monje et al. (2019).

TABLE 4 | Distribution according to the reason for explantation (*N* = 399 patients/521 implants).

	Patients	Implants
Peri-implantitis	242	323 (62%)
Primary failure	75	86 (16.51%)
Aseptic loosening	58	59 (11.32%)
Fracture	27	31 (5.95%)
Other reasons ^a	21	22 (4.22%)

Note: Some patients had multiple implants removed for different reasons.

^aOther reasons include malposition (*n* = 16), vestibular bone dehiscence (*n* = 2), aesthetics (*n* = 2), lack of function (*n* = 1), and pain (*n* = 1).

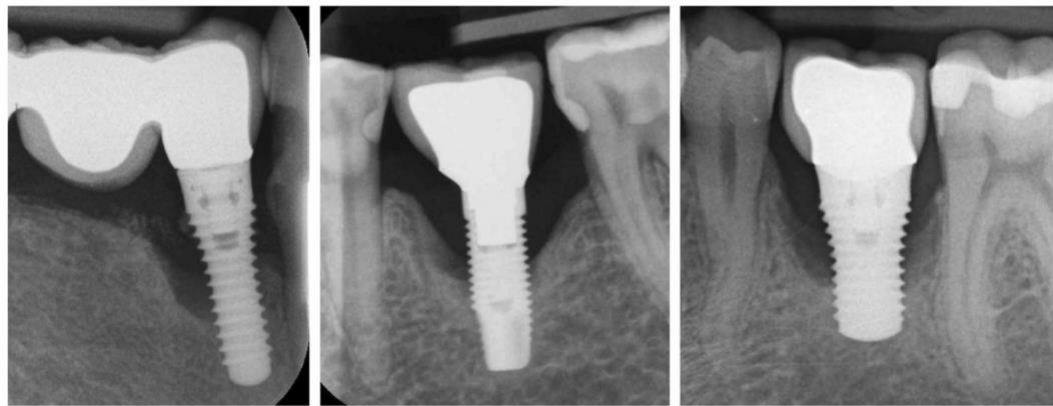


FIGURE 4 | Examples of explanted implants included in the study with approximately 60% of peri-implant bone loss.

TABLE 5 | Local characteristics of the explanted implants in the peri-implantitis group (calculated per implant).

	<i>N</i> (%)
Implant related	
Mean diameter (<i>N</i> =101)	4.21 mm
Diameter <4.0 mm	12 implants (11.88%)
Mean length (<i>N</i> =96)	9.75 mm
Length <10 mm	29 implants (30.21%)
Type of prosthesis (<i>N</i> =323)	
Fixed	231 (71.52%)
Removable	61 (18.89%)
Temporary	8 (2.48%)
No prosthesis	23 (7.12%)
Type of antagonist (<i>N</i> =277)	
Natural teeth	156 (48.30%)
Fixed implant-supported prosthesis	39 (12.07%)
Removable prosthesis	33 (10.22%)
Natural teeth + fixed implant-supported prosthesis	29 (8.98%)
Natural teeth + removable prosthesis	13 (4.02%)
No antagonist	7 (2.17%)

Data on tobacco consumption (*N*=182) revealed a high proportion of smokers (47.80%) with an average consumption of 13.89 cigarettes per day, ranging from 2 to 40. Additionally, 8.24% of the patients identified as former smokers; however, details regarding the duration since quitting and the quantity previously smoked were inconsistently reported.

3.4 | Other Analyses

3.4.1 | Follow-Up Visits

The proportion of patients with at least two follow-up appointments per year for scaling or supporting periodontal therapy was approximately 25% in the year preceding the explantation. This percentage dropped significantly, with only 10% of patients receiving consistent follow-up within the 5-year period before the explantation. Notably, 37% of patients had no follow-up visits during the year prior to the explantation.

3.4.2 | Relationship With Neighboring Anatomical Structures

The proximity of implants to neighboring anatomical structures was analyzed. Nearly 30% of implants were positioned within 1 mm of the maxillary sinus. Smaller proportions were

TABLE 6 | Periodontal and dental situation in the peri-implantitis group at the time of explantation (calculated per implant).

	<i>N</i> (%)	<i>SD</i>	Max	Min
Dental conditions at explantation				
Number of missing teeth (<i>N</i> =228)	14.41	8.24	28	1
Number of implants (<i>N</i> =232)	4.74	3.55	16	1
DMFT index (<i>N</i> =211)	22.75	5.80	28	3
Periodontal conditions at explantation				
Mean PPD (<i>N</i> =76)	3.84 mm	1.12	6.68	2.08
% PPD > 3 mm (<i>N</i> =76)	36.93%	21.83	78.07	2
% PPD > 5 mm (<i>N</i> =76)	17.33%	17.42	59.65	0
Bone loss at teeth (<i>N</i> =275)	254 (92.36%)			
% Root with bone loss at teeth (<i>N</i> =222)	45.66%	26.01	100	0
Bone loss/age (<i>N</i> =222)	0.74	0.46	2.33	0
History of periodontal treatment (<i>N</i> =218)	58 (17.96%)			

found for other structures: neighboring teeth or implants (12.38%), the inferior alveolar nerve (6.19%) and the nasal fossa (2.78%).

4 | Discussion

The present retrospective study aimed to identify the bone level at the time of explantation and to describe the profile of patients who underwent implant explantation. Implants were mainly explanted for peri-implantitis reasons, with a mean bone loss of 62.92% of the implant length, indicating severe peri-implantitis cases (Monje et al. 2019). These results agree with those from a recent study that reported a mean bone loss of 66.2% at explantation time (Wentorp et al. 2021) and with the current scientific evidence regarding the thresholds for implant explantation (Martin-Cabezas and Giannopoulou 2024). Furthermore, clinical studies have shown a decrease in successful treatment when bone loss around implants increases (de Waal et al. 2016). Hence, implants with more than 50% of bone loss at baseline increase the risk of implant failure after peri-implant surgical treatment by 20-fold compared to implants where bone loss was <25% of the length (Ravida et al. 2022). Moreover, in a

TABLE 7 | Fixed prosthesis in the peri-implantitis group (calculated per implant) related to number of crowns, abutments, and emergence profile.

Fixed prosthesis	n (%)
Number of patients (N=231 implants)	176 patients
Unitary	98 implants (42.42%)
Splinted	133 implants (57.58%)
Mesial	36 (27.07%)
Middle	27 (20.30%)
Distal	70 (52.63%)
Cantilever (N=182 implants)	
No	162 (89.01%)
Mesial	9 (4.95%)
Distal	11 (6.04%)
Microgap (N=225 implants)	56 (24.88%)
Number of abutments (N=230 implants)	
Mean number of abutments	2.06
1 abutment	100 (43.47%)
2 abutments	74 (32.17%)
3 abutments	28 (12.17%)
≥ 4 abutments	28 (12.17%)
Type of abutment (N=230 implants)	
Implants	202 (87.82%)
Implants + teeth	28 (12.17%)
Number of crowns (N=230 implants)	
Mean number of crowns	2.78
1 crown	100 (43.47%)
2 crowns	37 (16.09%)
3 crowns	39 (16.96%)
≥ 4 crowns	54 (23.48%)
Mesial emergence angle (N=215 implants)	30.41
30°	120 (55.81%)
≥ 30°	95 (44.19%)
Distal emergence angle (N=215 implants)	30.53
30°	121 (56.27%)
≥ 30°	94 (43.72%)
Mesial emergence profile (N=215 implants)	
Concave	59 (27.44%)

(Continues)

TABLE 7 | (Continued)

Fixed prosthesis	n (%)
Straight	136 (63.25%)
Convex	20 (9.30%)
Distal emergence profile (N=215 implants)	
Concave	70 (32.56%)
Straight	137 (63.72%)
Convex	8 (3.72%)
Crown:Implant ratio (N=226 implants)	0.91

recent retrospective analysis on the peri-implantitis surgical prognostic factors, implants with bone loss $\geq 60\%$ had a poorer survival prognosis when compared with implants with $<40\%$ (Romandini et al. 2024). These studies have shown an additional effect when simultaneous risk factors are present, such as smoking (de Waal et al. 2016), modified implant surface, or suppuration at baseline (Romandini et al. 2024).

Peri-implantitis was the main reason for explantation in this study, accounting for 62% of implants, followed by primary failure, aseptic loosening, and fractures. This aligns with the results of previous reports where explantations were performed in 64.5% (Gargallo-Albiol et al. 2021) and 59.9% of patients (Roy et al. 2020).

The population of this study presented a highly compromised oral health with a mean of 14.41 missing teeth and a DMFT score of 22.75 (DMFT index of 22.05 after exclusion of fully edentulous patients). Peri-implantitis has been associated with factors related to poor oral health factors such as the number of missing teeth, untreated interproximal caries or fillings adjacent to implants (Vilarrasa et al. 2021). Moreover, the population of our study had several implants per patient (4.74 ± 3.55). Peri-implantitis was found more frequently in patients with more than five implants (Passoni et al. 2014) and similar results were reported for patients with two or more implants (Gurgel et al. 2017).

Radiographic bone loss around teeth was present in 92.36% of the patients. This highlights the association between the history of periodontitis and peri-implantitis (Schwarz et al. 2018). Longitudinal studies have shown a high risk of peri-implantitis in patients with a history of periodontitis (Karoussis et al. 2003), especially in patients with severe periodontitis (Roccuzzo et al. 2012). Similar results were confirmed when periodontal charting was present, with most patients diagnosed with stages III (58%) and IV (40%) and grades B (50%) and C (50%). Moreover, the mean bone loss/age ratio was 0.74, which implies a high risk of periodontitis progression (Tonetti et al. 2018). Recently, stage IV and bone loss/age ratio >1 were associated with peri-implantitis occurrence in a longitudinal study (OR=41.29 and OR=8.87, respectively) (Romandini et al. 2025). Only 18% of the patients reported a history of periodontal treatment; taking

TABLE 8 | Removable prosthesis in the peri-implantitis group (calculated per implant).

Removable prosthesis	N (%)
Number of patients (N=59 implants) ^a	44 patients ^a
Explanted implants included on partial prosthesis	10 implants (16.95%)
Explanted implants included on complete prosthesis	49 implants (83.05%)
Indication for complete prosthesis	
Maxillary	35 (71.43%)
Mandibular	14 (28.57%)
Splinting	20 (33.90%)
Number of abutments (N=55 implants)	
Mean number of abutments of the reconstruction	3.55
1 abutment	2 (3.66%)
2 abutments	15 (27.27%)
3 abutments	9 (16.36%)
4 abutments	19 (34.55%)
≥ 5 abutments	10 (18.18%)
Complete maxillary with <4 implants (N=31)	6 (19.35%)
Complete mandibular with <2 implants (N=14)	0 (0%)
Type of abutments (N=57 implants)	
Implants	53 (92.98%)
Implants + teeth	4 (7.02%)

^aTwo medical files did not allow us to determine if the implant was an abutment of a complete or partial prosthesis. Therefore, analysis was performed on 44 patients/59 implants instead of 46 patients/61 implants.

into consideration the link between both diseases, this context could increase the proportion of implants explanted for peri-implantitis reasons.

Most frequently, explanted implants were restored by single-unit crowns (43%), followed by bridges with at least four crowns (23.48%). Long extension prostheses such as full arch bridges have been frequently associated with peri-implantitis (Romandini et al. 2025). In addition, maladaptation between the implant and the crown was radiographically evident at 24.88% of the explanted implants in our study. The gap can lead to a bacterial leakage, which would imply a peri-implant inflammatory reaction (Candotto et al. 2019).

No differences in the distribution of angle or emergence profile were found compared to other studies which reported angles $\geq 30^\circ$ (Yi et al. 2020) or $\geq 45^\circ$ (Corbella et al. 2023) as a risk of peri-implantitis. This fact could be due to the different types of

TABLE 9 | Systemic characteristics in the peri-implantitis group (calculated per patient).

	n (%/SD)
Systemic health (N=237)	
Healthy	133 (56.12%)
Diabetes	17 (7.17%)
Cardiovascular	35 (14.77%)
Hypertension	54 (22.78%)
Cholesterol	26 (10.97%)
Medications (N=234)	
Antidiabetic	16 (6.84%)
Antiplatelet	33 (14.10%)
Anticoagulant	13 (5.56%)
Antihypertensive	50 (21.37%)
Statin	24 (10.26%)
Antidepressant	16 (6.84%)
Smoking (N=182)	
Yes	87 (47.80%)
Former smoker	15 (8.24%)
Cigarettes per day (N=62)	13.89 (7.72)

implants in this study, including bone- and tissue-level implants. In fact, when Yi et al. (2020) analyzed only tissue-level implants, there was no significant association between the prosthetic factors and peri-implantitis (Yi et al. 2020). In our study, the lack of information regarding the type of connection prevented further analyses.

Only limited information exists in the literature regarding the link between peri-implantitis and systemic diseases with, however, contradictory results. For instance, a retrospective study found a significant impact on the early onset of peri-implantitis in patients with diabetes mellitus, osteoporosis, and hypertension; however, another study failed to find any association (Astolfi et al. 2022). The most frequently reported systemic disease in our study was hypertension, affecting 22.78% of the population, followed by cardiovascular diseases (14.77%). Similar values for hypertension (28.6%) have already been reported in patients with peri-implantitis (Anderson et al. 2020). Moreover, a recent meta-analysis concluded that the presence of cardiovascular diseases increases the risk of peri-implantitis (Chu et al. 2023).

Current evidence regarding the role of smoking as a risk factor for peri-implantitis is inconclusive (Schwarz et al. 2018). When data regarding smoking habits was available, 87 patients out of 182 reported smoking (47.80%), mainly heavy smokers (>10 cigarettes per day) (Ravida et al. 2020) with an average consumption of 13.89 cigarettes per day. Smoking has been associated with peri-implantitis incidence (OR=7.84) in a recent longitudinal prospective study (Romandini et al. 2025), and heavy

smokers have been associated with radiographic bone loss (Carral et al. 2021).

Finally, irregular follow-up or no follow-up was frequent during the years before the explantation. It has been reported that patients who are compliant with supporting therapy are less likely to develop peri-implantitis (Monje et al. 2017). In a study reassessing the peri-implant health of mucositis patients after 5 years, the lack of follow-up was associated with a 43.9% frequency of peri-implantitis compared to 18% of cases when the patient received at least 1 follow-up appointment per year (Costa et al. 2012).

The mean strength of our study is the sample size, which provides data with external validity. Moreover, a patient profile based on multiple local and systemic characteristics is described. In addition, including several private practices can shed light on the clinical practice trends, without focusing on a unique practitioner-based philosophy of treatment.

However, there are also some limitations inherent in the retrospective design of our study that should be clarified to correctly interpret the results. This design does not allow for determining the causal relationship of any of the analyzed factors/characteristics. For example, if most implant patients are over the age of 60, older individuals will naturally be overrepresented among those experiencing implant explantation. However, this overrepresentation does not necessarily imply a greater risk of implant loss associated with older age. Furthermore, different radiographic methods were available in the medical files, and information on the presence of systemic diseases was self-reported. Moreover, the morphology of the defect could not be assessed as only a few files included probing pocket depths or intra-surgical images. Consequently, only the type of defect and the severity could be analyzed. Other factors such as plaque level, keratinized mucosa dimensions, and previous peri-implant treatment were not consistently reported in the medical files, preventing their analysis.

Further studies should include control groups to determine the potential risk factors for implant loss. Moreover, studies should focus on the combined potential risk factors of peri-implantitis to determine a high-risk profile and to increase clinical efforts to maintain peri-implant health, thus avoiding explantation as ultimate treatment.

5 | Conclusion

Peri-implantitis was the most frequent cause of implant explantation. Explantations were typically performed once bone loss reached approximately 60% of the implant length. They were most often observed in advanced cases of peri-implantitis and in patients with high-risk profiles for periodontal breakdown, such as those with a history of periodontal disease or heavy smoking habits.

Author Contributions

Rodrigo Martín-Cabezas: conceptualization, data curation, formal analysis, visualization, writing – original draft, methodology,

investigation, writing – review and editing, validation, project administration. **Norbert Cionca:** writing – review and editing, validation. **Catherine Giannopoulou:** conceptualization, writing – review and editing, supervision, methodology, validation, project administration, formal analysis, visualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

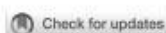
The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Residual bone level as a prognostic factor in the surgical treatment of peri-implantitis

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Peri-implantitis is a progressive inflammatory disease affecting the tissues surrounding dental implants and leading to bone loss. The severity of this disease is typically classified based on the depth of the bone defect or the percentage of bone loss around the implant. Marginal bone loss is a critical factor in the surgical management of peri-implantitis, as it can complicate access for implant decontamination and hinder efforts to stabilize the condition. In cases where bone loss exceeds 50% of the implant length, explantation is often recommended due to significantly reduced success rates after treatment. This narrative review seeks to examine the scientific evidence on marginal bone loss as a prognostic factor in the surgical treatment of peri-implantitis.

KEYWORDS

peri-implantitis, marginal bone loss, prognostic factor, surgical peri-implant treatment, treatment outcome, explantation

1 Introduction

Peri-implantitis is an inflammatory disease which affects the tissues surrounding dental implants, characterized clinically by increased probing pocket depth (PPD), bleeding on probing (BOP)/suppuration (SUP) and marginal bone loss (BL) beyond initial bone remodeling (1, 2). The disease progresses rapidly, following a non-linear accelerating pattern (3) that could lead to implant loss (4).

Different approaches for treating peri-implantitis have been proposed including non-surgical and surgical procedures (5). However, implants affected by peri-implantitis are considered to have doubtful prognosis (6) as despite the therapy, 30–50% of treated patients do not achieve complete disease resolution (7). These outcomes seem to be influenced by several factors, such as the implant surface (8), initial BL (9), baseline PPD (10), SUP at baseline (11) level of compliance, plaque control, diagnosis of severe periodontitis (12) and tobacco consumption (13).

The initial BL is related to the magnitude of the lesion. Hence, bone defects can be differentiated according to the BL pattern (intra-bony/supra-bony), the number of residual bone walls (dehiscence, 2-, 3-walls or circumferential) and the severity of the bone defect (relative BL related to the total implant length or the intra-surgical defect depth in millimeters) (14–16).

When BL progresses, a significant portion of the implant surface becomes exposed to the oral cavity, making it susceptible to biofilm contamination. Removal of the biofilm can be particularly challenging, especially on implants with rough or modified surfaces (11, 13). In these cases, peri-implant surgery allows better access to the implant surface in order to achieve optimal decontamination (17). However, depending on the amount

of initial BL and its morphology, the instrumentation of the implant surface is hindered, with the apical part of the implant becoming less accessible and more difficult to decontaminate (18). Moreover, while suprabony defects allow better instrumentation, intrabony defects could hamper the possibilities for thorough decontamination (19, 20) (Figure 1).

When treating deep defects by resective approaches, the apical repositioning of the flap can be impossible due to the remaining vestibulum height (11). On the other hand, it is not likely that deep bone defects remain self-containing with preservation of the bone walls, which could impact the prognosis for regenerative approaches (21). In addition, BL can result in unfavorable configurations of hard and soft tissues related to the neighboring teeth, that could also hinder optimal cleaning (13). As recently supported by Monje et al. (2023), it can also cause proximal loss of periodontal support for these teeth (22). Finally, after the surgical treatment, a reduced bone level has also been associated with recurrence or progression of the disease (23).

Implant removal can be the most viable option in some severe peri-implantitis cases where the surgical approach has low predictability (24). However, there is no universally accepted threshold beyond which implant preservation becomes impossible (25, 26).

In consequence, the marginal bone level prior to the surgery becomes a key factor in the decision-making process. The aim of

this narrative review is to analyze the scientific evidence related to the initial BL in peri-implantitis as prognostic factor prior to surgical treatment.

2 Marginal bone loss assessment and classification

Bone remodeling is a physiological process that occurs during the first months of function after prosthesis delivery (2). Historically, different thresholds were proposed to clinically distinguish the physiological bone remodeling from the pathological bone loss (27). Consequently, these differences had affected the reported prevalence of the disease (28). It is accepted that bone remodeling should be limited to 2 mm after prosthesis delivery, and that beyond this threshold, it should be considered as pathological (2). However, these criteria are older and nowadays we should aim to prevent excess remodeling (29).

The marginal bone level is a primary diagnostic factor for peri-implantitis, with a radiographic sensitivity threshold for detecting bone changes of 1.0 mm (30). When initial data is not available to assess the limit of bone remodeling, a 3-mm BL threshold has been proposed for the diagnosis of peri-implantitis in conjunction with the clinical measurements (1, 2).

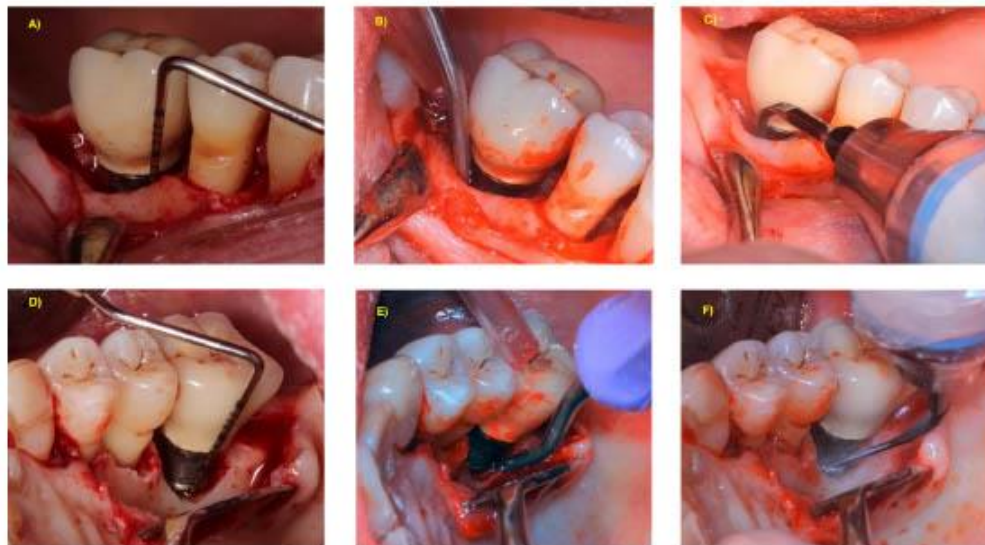


FIGURE 1
(A–C) Access for implant surface decontamination of deep intrabony defect. (D–F) Access for implant surface decontamination of deep suprabony defect.

The long-cone parallel intraoral radiographic projection is recommended for peri-implant bone evaluation which allows to assess crestal bone changes longitudinally (27). The evaluation of marginal BL on radiographs has shown a positive linear correlation between mesial and distal marginal bone levels and the BL evaluated clinically during surgery (30, 31); however, the radiographic measurements resulted frequently in underestimation of the real bone loss assessed after flap elevation (mean 0.7 and 0.6 mm for mesial and distal, respectively) (30).

Although this radiographic technique is considered the gold-standard, it presents an important limitation related to the lack of information on the vestibular and lingual bone walls. When analyzing the residual walls, the morphology of the bone defect was correlated to the buccolingual crestal width, and 4-walls defects were found in broader crests (31). It is important to consider that the vertical component of the BL at buccal and lingual aspects was found to be statistically deeper at 4-walls defects than at 2-walls defects (31). Other authors have analyzed three dimensional assessment of peri-implant BL, however, scatter and artefacts may play a role in reducing the quality of the CBCT imaging, limiting the indication of this radiographic method for the evaluation of peri-implant defects (32).

Based on the literature, multiple classifications have been developed to categorize peri-implant BL (14–16, 33). While Schwarz et al, included only the type of bone loss (suprabony/infrabony) and the residual bone walls (16), other studies proposed the classification of peri-implantitis based on the severity of the BL as follows: Early/Slight, Moderate or Advanced peri-implantitis (BL <25%; 25–50%; >50% of the implant length, respectively) (14, 15, 33). Moreover, the position of the implant in relation to the alveolar crest has recently been included in the classification (BL due to implant malposition or ridge defects). This tridimensional consideration may have prognostic value on determining the best choice of the surgical approach, as for example, regenerative approaches can be limited if implants are placed outside the bone housing (33).

3 Initial bone loss and prognosis

Several studies (7–9, 11–13, 34, 35) have analyzed the negative impact of the initial marginal bone levels on the surgical peri-implant outcomes in terms of healing, stabilization or implant failure. All these studies reported an inverse association between the initial severity of the disease and the therapeutic effectiveness. However, the degree of severity is not consistently reported across the studies, and it is classified heterogeneously. Table 1 summarizes the main findings of these studies.

3.1 Access flap and resective surgery

Serino et al, conducted a retrospective clinical study on a cohort of patients treated for peri-implantitis by resective surgery for elimination of angular bony defects. The study included implants with BL ≥ 2 mm with a mean BL of 5.5 mm, and single implant

restorations were excluded. Subjects were recalled every 6 months for re-evaluation and maintenance care. Sub-gingival scaling at implants presenting with residual pockets was performed during these sessions. At ten-years re-evaluation, 66.6% of implants presenting with deep pockets and inflammation at baseline, remained stable throughout the follow-up period, whereas 29% of the cohort showed further bone loss progression. The authors concluded that the peri-implantitis therapy combined with a high standard of oral hygiene and a 6-month recall program was effective for most of the implants. They also reported that advanced BL before treatment was associated with residual pockets following peri-implant surgery which may be considered a risk indicator for disease progression. As clinical recommendation, authors suggested that the explantation of implants with advanced BL may be the choice of treatment, when a regenerative approach is not feasible and when the prosthetic reconstruction is not compromised. However, it is important to mention that while, this report described “advanced/pronounced/substantial” BL, no threshold was proposed in terms of millimeters or percentage of implant length (7).

The prognostic value of BL in millimeters was evaluated in other studies. In a prospective study including 86 implants treated for peri-implantitis by resective surgery, the outcomes of the therapy were classified according to the initial BL. At 2 years follow-up, 76% of the implants with initial BL between 2 and 4 mm were classified as healthy (PPD < 4 mm without bleeding, suppuration of further BL). In the group with initial BL between 5 and 6 mm, 55% of implants were healthy at the follow-up, while only 22% of implants with an initial BL of ≥ 7 mm was classified as healthy. Furthermore, explantations were performed exclusively in the group with initial BL of ≥ 7 mm, with a total of 7 explantations (39%) at 6 months (34). It is important to mention that the treatment of advanced bone defects and deep intrabony defects was limited to bone re-contouring with a resective approach. Thus, for the management of deep intrabony defects, augmentative or combined surgical therapy should be considered (36).

In the same way, the presence of initial BL > 7 mm has been associated to persisting peri-implant inflammation after resective surgery in terms of SUP, BOP or the combination of PPD + BOP (4 mm or 6 mm) (11). In this study with short-term follow-up (6 months), all implants were still present after the therapy. Thus, marginal BL was related to the treatment efficacy, in terms of clinical parameters (SUP, BOP and PPD reduction).

The study of de Waal et al. (2016) was based on two previously conducted randomized controlled trials involving treatment of peri-implantitis with a resective approach. Both trials were conducted by the same group (13). The authors reported that the mean BL at baseline (4.1 ± 1.6 mm) was associated with failure of peri-implantitis treatment after 1 year (OR = 1.46; 95% CI: 1.0–2.1; $p = 0.043$). While implants with a mean initial BL of 1–3 mm showed successful treatment outcome (more than 70% of implants), those with 7–9 mm initial BL showed lower success rates (25% of implants). Among other factors, smoking was associated with poorer treatment outcome. Moreover, the authors highlighted that there is a learning curve for the surgical

TABLE 1 Studies assessing the association between severity of marginal bone loss and therapeutic outcome.

Study Country	N	Type of study	BL classification	Type of surgery	Follow-up	Outcome
Serino & Turri (34), Sweden	31 P/168 I	Prospective case series	2–4 mm 5–6 mm >7 mm	Resective	2 years	Resolution of the disease depended on initial BL. Persistence of the disease is most frequent when initial BL ≥ 5 mm
Lagervall & Jansson (12), Sweden	150 P/382 I	Retrospective case series	No BL BL $\leq 1/3$ BL $> 1/3 \leq 2/3$ BL $> 2/3$	Resective or regenerative (xenograft + collagenous membrane)	26 $\pm$ 20 months	BL $> 1/3$ of the length of the implant reduced the effectiveness or the therapy
de Waal et al. (13), Netherlands	74 P/106 I	Retrospective analysis of two RCT	1–3 mm 3–5 mm 5–7 mm 7–9 mm	Resective	1 year	BL at baseline were associated with decreasing success rates
Koldstad et al. (28), Norway	45 P/143 I	Prospective case series	2–3 mm >3 mm >5 mm >7 mm	Resective	6 months	BL > 7 mm was associated with persisting disease (SUP, BOP, PPD > 4 mm + BOP or PPD > 6 mm + BOP)
Serino et al. (7), Sweden	At 2 years: 39 P/208 I At 10 years: 18 P/85 I	Retrospective case series	NR	Resective	2 years 10 years	The effectiveness of the treatment was inversely correlated to the initial bone loss
Lee et al. (35), Korea	45 P/92 I	Retrospective case series	<3 mm 3 mm ≥ 4 mm	Non-surgical or access flap	6.4 $\pm$ 2.7 years	Implants with BL < 3 mm showed better outcome than those with BL ≥ 4 mm. OR = 5.15; 95% CI: 1.20–22.07, $p = 0.027$.
Ravida et al. (9), USA	80 P/121 I	Retrospective case series	Mild $< 25\%$ Moderate 25%–50% Severe $> 50\%$	Resective or regenerative (allograft and/or xenograft + collagenous membrane)	42.6 $\pm$ 26.3 months	BL $> 50\%$ length increased 20 times the risk of implant failure, compared to implants with BL $< 25\%$. OR = 20.2; 95% CI: 2.42–169.6; $p = 0.006$.
Romandini et al. (8)	149 P/267 I	Retrospective analysis of one RCT and one retrospective case series	BL $< 40\%$ BL $\geq 40\%$ $< 60\%$ BL $\geq 60\%$	Access flap or resective	7.0 $\pm$ 3.6 years	BL $\geq 60\%$ at baseline was a predictor of implant failure

N, population; BL, bone loss; P, participants; I, implants; SUP, suppuration; BOP, bleeding on probing; PPD, probing pocket depth; NR, no reported; OR, odds ratio; CI, confidence interval.

treatment of peri-implantitis as the amount of experience of the surgical team was significantly associated to treatment success.

It is important to highlight that reporting marginal BL in millimeters has different impact depending also on the initial implant length. In terms of relative BL, Romandini et al. (8), have compared different percentages at baseline as prognostic factors in access flap and resective approaches in the treatment of peri-implantitis. The authors classified relative BL in three categories: $<40\%$; $40\text{--}60\%$; and $\geq 60\%$. Implants with BL $\geq 60\%$ presented the highest risk for implant loss. Interestingly, implants with BL between 40 and 60% did not show significant risk for implant failure compared to implants with BL $< 40\%$ ($p = 0.205$).

3.2 Reconstructive surgery

For reconstructive surgery, studies analyzed mainly the impact of the defect configuration and residual bone walls on the radiographic bone fill. However, the severity of the defect on treatment effectiveness was not always reported (21). The impact of defect depth has shown controversial results regarding radiographic bone fill. Aghazadeh et al. analyzed the use of

resorbable membranes in conjunction with autogenous bone or xenograft, at different types of defect morphology. The authors reported significant radiographic defect fill at the deepest bone defects; a correlation between mesial and distal initial defect depth measured at the time of surgery, and the radiographic outcome was observed (31). On the contrary, another study did not find a correlation between radiographic bone gain and initial marginal bone level (37). However, when analyzing disease resolution with reconstructive surgery in relation to the baseline marginal bone level, slight BL ($<25\%$ of the implant length) resulted in better outcomes than the moderate (25%–50% of the implant length) or advanced cases ($>50\%$ of the implant length). The success rates were found to be 84.6%, 75% and 71.4%, respectively (37).

The impact of defect morphology has also been analyzed with however conflicting results. While some studies showed that the 4-walls lesion has better reconstructive potential compared to the 2- and 3-walls defects (21, 31), other studies failed to find an association between defect configuration and treatment outcomes (10, 38). Moreover, the defect angle showed a correlation with the radiographic bone gain, with a better defect fill in cases of narrow angles ($<40^\circ$) (37).

3.3 All approaches analyzed together

Some studies pooled together the resective and regenerative approach. Lagervall and Jansson (12) analyzed retrospectively the prognostic factors for treatment success, including non-surgical, surgical resective and regenerative approaches. BL was classified according to the different thirds of the implant length, in degree 1, 2 or 3; severe peri-implantitis was considered when BL was greater than one third of the implant length (degree 2 and 3). Non-surgical therapy was most frequent in patients with BL less than one third of the implant length (degree 1), while defects of degree 2 and 3 were most frequently treated by surgery. The regenerative approach was most frequently performed in severe cases rather than those of degree 1, which could explain the lower success rates of this approach. Explantations were only performed in four patients presenting defects with degree 3. The severity of the lesion reduced the success of the therapy. Implants with BL degree 1 were less prone to have peri-implantitis at the last reevaluation after treatment, compared to those with degree 2 and 3 [Odds Ratio (OR) = 6.5; 95% CI: 1.4–30; $p < 0.05$]. However, degrees 2 and 3 were not analyzed independently, thus not allowing to differentiate between the middle and the apical thirds of the initial BL in terms of prognosis.

Recently, Ravidá et al. (9) included also both types of surgery to assess the baseline factors that could influence the outcome. The authors reported that increased BL and implant location (anterior) were associated with implant failure. BL between 25% and 50% of the length of the implant increased the risk of failure by 15 times (OR = 15.2; 95% CI: 2.06–112.7; $p = 0.008$); and BL > 50% increased the risk by 20 times (OR = 20.2; 95% CI: 2.42–169.6; $p = 0.006$). The risk in both cases, was significantly higher compared to implants with BL < 25%. The authors further analyzed independently the two approaches and BL was confirmed to be a prognostic factor for implant failure in both cases. At the end of the study, 39% of implants with > 50% of BL were removed, 78.6% did not reach the treatment success criteria (PPD ≤ 5 mm, absence of BOP and bone loss ≤ 0.5 mm), and only 21% belonging to the group with initial BL of > 50% showed clinical signs of health.

4 Discussion

The baseline BL prior to treatment can influence the choice of the peri-implant surgical approach (24) and the treatment outcome (13). Several studies have shown low success rates when initial BL is advanced (8, 9, 13). BL is inversely correlated to the effectiveness of therapy: the highest the baseline BL, the lowest the therapeutic success rate, independently of the surgical approach (resective or regenerative) (7–9, 13). Moreover, it has been shown that each additional 1-mm of BL prior to the surgical treatment, increases the risk of future implant failure by 65% (39).

The classification of BL used in the studies is heterogenous, not allowing to establish a cutoff point for explantation. The results of the analyzed studies depend on the reference point for the statistics

and conclusions should be interpreted carefully. Thus, when the reference point was stated at BL < 40, the statistic threshold for implant failure was BL > 60% (8); however, when the reference was < 25%, the risk for failure increased statistically even for BL 25%–50% (9). Similar results occurred when the reference point was 33% of the implant length, with higher risk for BL up to the middle third (12). The differences between studies can be in part due to the disparities on “advanced BL” definition. Thus, it is important to report consistently the severity of the lesion, keeping the term “advanced” for BL > 50% (14, 15, 33).

When BL is restricted to one third of the implant length, non-surgical therapy may, in some cases, resolve the disease without further need of surgical treatment (12); however, surgical approaches in those situations, have shown better outcomes with success rates up to 70% (13, 34, 37). More factors other than apical BL extent such as defect configuration or soft tissue quality may influence the therapeutic choice and outcome (9, 31, 39).

It is recognized that for the treatment of the majority of advanced peri-implant lesions, surgical approaches are often needed (40). However, when the marginal BL reaches the apical region of the implant, the prognosis becomes hopeless and explantation should be considered (6). In fact, the stabilization of implants with BL > two thirds of the length is unpredictable (8).

From a clinical point of view, the diagnosis of peri-implantitis when baseline data is not available requires a PD ≥ 6 mm + BOP/SUP + marginal BL ≥ 3 mm (1). Considering a standard implant (10 mm length), BL would be almost at the middle third. When the implant BL affects the middle third of the implant (33%–66%), the decision between peri-implant surgery or explantation becomes more sensitive. When BL is in the middle third, studies have shown worst prognosis compared to those presenting BL < 25%–33% of the length (9, 12), however another study did not found significant differences between BL < 60% and BL < 40% (8). Thresholds of BL > 50%–60% of the implant appear to be aligned with the clinical reality. Furthermore, a study analyzing at which BL level implants affected by peri-implantitis were explanted, concluded that the practitioners performed explantations when implants presented a mean BL of 66.5% (25).

In this context, BL exceeding 50% of the implant length has been proposed as threshold for explantation (4, 24). However, the stabilization of implants with advanced BL is also achievable, thus explantation based only on bone levels could result in the loss of opportunity for some patients to maintain their implants (Figure 2). Moreover, although the success rates of implants placed in early failed implants sites are high (96% compared to 98% for implants placed for the first time), the new implants had significantly higher risk of failure (41). Hence, “rescue therapy” can be performed in advanced cases, depending also on patient desires (24) as the surgical treatment performed in implants with BL > 50% has shown 21% success rate (9), thus avoiding the sequela from implant explantation (42). In those cases a thorough examination of the risk indicators is important for establishing a pretherapeutic prognosis, such as hard and soft tissue deficiencies, tobacco consumption, medical condition, implant surface or cleanability of the prosthetic restoration (6, 8, 13). In fact, smoking is associated to advanced BL (13), residual pockets



FIGURE 2

1-year stabilization of advanced peri-implantitis showing BL > 50% of implant length with combined surgical approach (reconstructive + implantoplasty). (A) Baseline clinical situation with PPD = 8 mm and BOP; (B) periapical x-ray showing intrabony bone loss up to the apical third of the implant; (C) clinical situation at 1-year post-operative follow-up with PPD = 4 mm without BOP; (D) periapical x-ray showing complete bone filling of the intrabony component.

after surgical peri-implant treatment, and is considered as a prognostic indicator in treatment outcome (13). Thus, smoking cessation is advised in both pre- and post-surgical care (10). In addition, technical factors that may influence our treatment choice, such as the risk of potential damage of the neighboring anatomical structures (maxillary sinus or inferior alveolar nerve) or the prosthetic implication of the implant, should be considered.

Further research for development and validation of composite models for peri-implant prognosis is needed. These models should combine several risk factors to allow the practitioner to decide the best option according to the risk classification. In our knowledge, only one model is nowadays available based on Nobel Biocare implants (43). This model considers that BL of more than one third of implant length (>33%) and combined with factors, such as history of periodontitis, early disease development (<4 years of function) and implant length >13 mm, as having unfavorable prognosis. Unfortunately, this model cannot distinguish finer degrees of BL, highlighting the need for further research to support practitioners in making evidence-based decisions regarding explantation or peri-implant surgery.

In conclusion, several limiting factors pertain to the early diagnosis of peri-implantitis and its accuracy, such as the low sensitivity of intra-oral radiographs for detecting bone changes of approximately 1.0 mm (30), the low sensitivity of peri-implantitis classification for early-stage detection of the disease in the absence of baseline data (44), the rapid progression of the disease (3), and the prognostic significance of initial marginal BL (13). This situation highlights the importance of baseline documentation following implant placement for the early diagnosis and treatment of peri-implantitis.

5 Conclusion

BL is a key pretherapeutic prognostic factor for peri-implant surgical treatment outcome, as treatment effectiveness is inversely correlated with initial bone levels. In advanced cases, where BL exceeds 50–60% the success rates decrease significantly, thus the decision for explantation should be considered.

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DISCUSSION

Parmi les 521 implants explantés dans l'étude, la principale cause d'explantation a été la péri-implantite, représentant 62 % des cas, suivie par l'échec primaire, la perte aseptique et les fractures. Ce résultat est similaire aux résultats obtenus dans une autre étude rétrospective où les péri-implantites ont impliqué 64,5 % des explantations (Gargallo-Albiol et al., 2021). L'interprétation de ces résultats doit tenir compte de la faible incidence d'autres causes d'explantation, comme les fractures (2,3 %) (Stoichkov & Kirov, 2018), face à la prévalence des péri-implantites (19,83 % au niveau des patients et 9.25 % au niveau des implants) (Lee et al., 2017). De même, le 88,9 % des implants atteints de péri-implantites s'est développé sans symptômes pour le patient et ont un faible impact sur la perception de la qualité de vie des patients (Romandini et al., 2021). Ce fait, couplé au taux de progression rapide de la pathologie (Derks et al., 2016), implique fréquemment un diagnostic tardif et un traitement moins prévisible.

En effet lors de la classification des implants explantés dans notre étude selon la sévérité du défaut osseux (Monje et al., 2019), 80,42 % des cas étaient des péri-implantites sévères et seulement un cas présentait une perte osseuse inférieure à 25 %. Ce fait peut s'expliquer par le meilleur pronostic des cas débutants traités avec le traitement péri-implantaire (de Waal et al., 2016). En effet, pour les cas avec une perte osseuse inférieure à un tiers de la longueur, le traitement non chirurgical peut suffire à la guérison de la pathologie (Lagervall & Jansson, 2013).

Notre étude a montré que le seuil à partir duquel l'explantation des implants atteints de péri-implantite a été réalisée est de 62,92 %. Ce niveau de perte osseuse est en accord avec les résultats obtenus dans une autre étude rétrospective ayant montré un seuil de 66,2 % (Wentorp et al., 2021). Les études cliniques analysant les résultats du traitement chirurgical ont mis en évidence que plus la lésion initiale est profonde, plus le pronostic de guérison est faible (de Waal et al., 2016) et l'explantation a été conseillée pour les implants avec une perte osseuse qui dépasse 50 % (Sinjab et al., 2018; Zimmer et al., 2025). Ainsi, pour ces cas de péri-implantite avec une perte supérieure ou égale à 60 % de la longueur de l'implant, le risque de perte implantaire est 4,21 fois plus important comparé aux implants avec une perte osseuse inférieure à 40 % (Romandini et al., 2024).

Bien que les implants avec une perte osseuse sévère aient un pronostic imprévisible (Romandini et al., 2024), cela ne signifie pas que leur conservation soit toujours impossible. En effet, le traitement des implants avec une perte osseuse supérieure à 50 % a montré une stabilisation dans 21 % des cas (Ravida et al., 2022). D'autres facteurs tels que le tabac (de Waal et al., 2016), la surface implantaire (Romandini et al., 2024) ou l'observance du suivi (Lagervall & Jansson, 2013) ont également montré un taux de réussite plus faible dans le traitement implantaire et peuvent jouer un rôle additionnel dans la réussite du traitement (de Waal et al., 2016). Les caractéristiques locales et systémiques des patients de notre étude ont été analysées :

Caractéristiques bucco-dentaires.

L'état de la denture global de notre étude a montré un indice de dents cariés, absents ou traités (DMFT) élevé (index = 22,75). Plusieurs associations entre le niveau de santé bucco-dentaire et les péri-implantites ont été rapportées dans la littérature (Vilarrasa et al., 2021). Plus particulièrement, les patients avec un antécédent de parodontite ont montré un risque de développer une péri-implantite plus important que les patients avec un parodonte sain (Roccuzzo et al., 2012; Schwarz et al., 2018). Le sondage parodontal n'était pas disponible systématiquement dans le dossier des patients inclus dans notre étude et certains facteurs tels que la plaque, la profondeur de sondage ou l'inflammation n'ont pas pu être analysés.

Toutefois, la perte d'attache au niveau dentaire a pu être mise en évidence par l'analyse radiographique, et une perte osseuse a été identifiée chez 92,36 % des patients, atteignant en moyenne 45,66 % de la longueur de la racine. Cette situation met en évidence une susceptibilité parodontale parmi les patients étudiés.

Quant aux restaurations prothétiques, la majorité ont été des prothèses fixes (71,52 %), notamment unitaires, avec une distribution similaire, indépendamment du profil d'émergence. Ceci pourrait être dû à l'inclusion dans l'étude de différents types d'implants « bone/tissue level » (Yi et al., 2020).

Caractéristiques systémiques et consommation de tabac

Lors de l'analyse des données de santé globale, seulement 56,12 % ont montré un historique sans rien à signaler. D'autres pathologies comme l'hypertension ou les maladies cardiovasculaires ont été prévalentes dans cette population (22,78 % et 14,77 %, respectivement), ce qui a été lié à la prise de médicaments antihypertenseurs (21,37 %), antiplaquettaires (14,10 %) ou anticoagulants (5,56%).

De plus, la prévalence de fumeurs était élevée avec 47,80 % de fumeurs et 8,24 % d'ancien fumeurs, et une consommation moyenne de 13,89 cigarettes par jour. L'évidence pour l'association entre le tabac et les péri-implantites est considérée à ce jour inconcluante, avec des résultats variables dans la littérature (Schwarz et al., 2018). Cependant, le lien entre la consommation de tabac et la parodontite ainsi que la perte dentaire est bien établi (Chaffee et al., 2021), ce qui pourrait constituer un facteur indirect.

Suivi dentaire et profil des patients

Ces conditions de santé bucco-dentaire ont été accompagnées d'un faible suivi dentaire. Par exemple, seulement 18 % avaient suivi un traitement parodontal préalable, et le suivi chez l'hygiéniste dentaire dans les années précédant l'explantation était irrégulier ou inexistant.

Ainsi, les données actuelles montrent que le manque de suivi régulier a été associé aux péri-implantites (Schwarz et al., 2018) et à un taux de conversion de mucosite à péri-implantite plus élevé (Costa et al., 2012).

Cette étude a montré que, parmi les patients ayant subi une explantation, la santé bucco-dentaire et systémique était fréquemment compromise. De plus, les patients ont eu un historique de suivi dentaire insuffisant.

La littérature a montré que 74 % des patients pensent qu'un implant est à vie, avec peu d'informations concernant les pathologies péri-implantaires et leurs facteurs de risque (Insua et al., 2017). Ce fait souligne l'importance de la prévention primaire, du suivi et de l'information auprès du patient avant le traitement implantaire (Carra et al., 2023).

Forces et limitations

Pour mieux comprendre la portée et les limites de cette étude, nous avons évalué les forces méthodologiques ainsi que les contraintes rencontrées. Cette étude a analysé une cohorte large avec 242 patients et 323 implants explantés pour des péri-implantites inclus, avec plusieurs données relatives au patient, à l'implant, à la reconstruction prothétique, au niveau de santé buccale et systémique et à la consommation de tabac qui ont été analysées.

L'information obtenue dans cette étude a une validité externe et pourrait permettre de cibler un profil de patient susceptible de subir un échec implantaire tardif et pour lequel les efforts en matière de prévention et d'information auprès du patient devraient être renforcés.

Cependant, les résultats de cette recherche doivent être interprétés avec précaution, car certaines limitations sont présentes. En effet, une étude rétrospective sans groupe contrôle ne permet pas de mettre en évidence l'association entre deux pathologies, empêchant d'identifier des facteurs de risque.

De plus, étant une étude basée sur le dossier du patient, les données médicales du patient étaient auto-déclarées, certaines manquaient, ce qui a empêché l'analyse de quelques facteurs importants comme la profondeur de sondage ou le type de surface implantaire.

Malgré la description dans cette étude des caractéristiques des patients ayant subi une explantation, la réalisation de futures études comparatives est nécessaire pour déterminer des facteurs de risque.

Par ailleurs, l'utilisation de seuils et de définitions homogènes de sévérité dans la recherche peut faciliter l'interprétation et la comparaison des résultats.

CONCLUSION

En résumé, les résultats de notre étude mettent en évidence que les chances de stabilisation pour les implants avec une perte osseuse supérieure à 50 % de la longueur de l'implant sont faibles et le traitement devient imprévisible. Ainsi, un seuil d'explantation entre 50 % et 60 % semble être observé tant dans la pratique clinique que dans la littérature scientifique.

Les explantations ont été réalisées chez des patients avec un faible niveau de santé bucco-dentaire et systémique, ainsi qu'une consommation tabagique élevée et un suivi insuffisant.

Ces informations soulignent l'importance de l'information auprès du patient, de la prise en charge et de l'établissement d'un suivi dentaire chez les patients candidats aux restaurations implanto-portées.

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